

 **B.TECH 1ST YEAR · 1ST SEMESTER**

Engineering Mathematics

Partial Differentiation & Applications

UNIT IV — Complete Formula Sheet

Every formula · All conditions · Key theorems · Exam tips ·
Flashcards · Quick revision

Last-Day Revision Weapon · Exam-Ready · Memory-Optimized

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1. FUNCTIONS OF SEVERAL VARIABLES

Definition

GENERAL FORM

$z = f(x, y)$ – two variables

$w = f(x, y, z)$ – three variables

$u = f(x_1, x_2, \dots, x_n)$ – n variables

VARIABLES

$x, y \rightarrow$ Independent variables

$z \rightarrow$ Dependent variable

Domain & Level Curves

DOMAIN

Set of all (x, y) for which $f(x, y)$ is defined and real

LEVEL CURVE / CONTOUR

$f(x, y) = c$ (constant)

Set of points where f has same value c

2. LIMIT, CONTINUITY & DIFFERENTIABILITY

2.1 Limit of $f(x, y)$

DEFINITION

$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$

$f(x, y)$ approaches L as $(x, y) \rightarrow (a, b)$ along ALL paths

KEY: Limit Does NOT Exist If:

Two different paths give two different limit values $\rightarrow$ limit DNE

2.2 Continuity Conditions

$F(x, y)$ IS CONTINUOUS AT (a, b) IF ALL THREE HOLD:

① $f(a, b)$ is defined

② $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists

③ $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$

2.3 Differentiability

CONDITION

f is differentiable at (a, b) if partial derivatives f_x and f_y exist AND are continuous near (a, b)

★ Differentiability $\implies$ Continuity, but Continuity $\nRightarrow$ Differentiability

★ Existence of partial derivatives $\nRightarrow$ Differentiability

3. PARTIAL DERIVATIVES

First Order Partial

$\partial z / \partial x$ — PARTIAL W.R.T. x

Notation Forms

Notation

Meaning

$$\frac{\partial f}{\partial x} = f_x = \lim_{h \rightarrow 0} \frac{[f(x+h, y) - f(x, y)]}{h}$$

(treat y as constant)

$\frac{\partial z}{\partial y}$ — PARTIAL W.R.T. Y

$$\frac{\partial f}{\partial y} = f_y = \lim_{k \rightarrow 0} \frac{[f(x, y+k) - f(x, y)]}{k}$$

(treat x as constant)

$\frac{\partial f}{\partial x}$	Leibniz
f_x or f_1	Subscript
$D_x f$	Operator
$\frac{\partial z}{\partial x}$	Dependent variable

Second & Higher Order Partial Derivatives

Symbol	Full Form	Meaning
f_{xx}	$\frac{\partial^2 f}{\partial x^2}$	Differentiate f_x w.r.t. x
f_{yy}	$\frac{\partial^2 f}{\partial y^2}$	Differentiate f_y w.r.t. y
f_{xy}	$\frac{\partial^2 f}{\partial y \partial x}$	Differentiate f_x w.r.t. y
f_{yx}	$\frac{\partial^2 f}{\partial x \partial y}$	Differentiate f_y w.r.t. x

CLAIRAUT'S / SCHWARZ'S THEOREM (MIXED PARTIALS)

If f_{xy} and f_{yx} are **continuous**, then: $f_{xy} = f_{yx}$ (Mixed partials are equal)

4. TOTAL DERIVATIVE

4.1 Total Differential

$$dz = (\partial z / \partial x) dx + (\partial z / \partial y) dy$$

FOR $W = F(X, Y, Z)$

$$dw = (\partial w / \partial x) dx + (\partial w / \partial y) dy + (\partial w / \partial z) dz$$

★ Meaning: Combined infinitesimal change due to changes in all independent variables

4.2 Total Derivative (when variables depend on t)

$$dz/dt = (\partial z / \partial x)(dx/dt) + (\partial z / \partial y)(dy/dt)$$

★ Used when $z = f(x, y)$ and $x = x(t)$, $y = y(t)$ all depend on single parameter t

5. CHAIN RULE (MULTIVARIABLE)

Case	Function	Chain Rule Formula
Case 1 One parameter	$z = f(x, y)$ $x = x(t)$, $y = y(t)$	$dz/dt = (\partial z / \partial x)(dx/dt) + (\partial z / \partial y)(dy/dt)$
Case 2 Two parameters	$z = f(x, y)$ $x = x(s, t)$, $y = y(s, t)$	$\partial z / \partial s = (\partial z / \partial x)(\partial x / \partial s) + (\partial z / \partial y)(\partial y / \partial s)$ $\partial z / \partial t = (\partial z / \partial x)(\partial x / \partial t) + (\partial z / \partial y)(\partial y / \partial t)$
Case 3 Three variables	$w = f(x, y, z)$ $x, y, z =$ functions of t	$dw/dt = (\partial w / \partial x)(dx/dt) + (\partial w / \partial y)(dy/dt) + (\partial w / \partial z)(dz/dt)$
Implicit $F(x, y) = 0$	$F(x, y) = 0$	$dy/dx = -F_x / F_y \quad (F_y \neq 0)$
Implicit $F(x, y, z) = 0$	$F(x, y, z) = 0$	$\partial z / \partial x = -F_x / F_z, \quad \partial z / \partial y = -F_y / F_z$

CHAIN RULE MEMORY TRICK

"Each path to the final variable contributes one term. Multiply along each path, add all paths."

★ 6. EULER'S THEOREM (HOMOGENEOUS FUNCTIONS)

DEFINITION: HOMOGENEOUS FUNCTION OF DEGREE N

$$f(tx, ty) = t^N \cdot f(x, y) \quad \text{for all } t$$

EULER'S THEOREM

SECOND ORDER EXTENSION

$$x \cdot f_x + y \cdot f_y = n \cdot f$$

$$x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1)f$$

THREE VARIABLES: $U = F(X,Y,Z)$, DEGREE N

$$x \cdot U_x + y \cdot U_y + z \cdot U_z = n \cdot U$$

Example Function	Degree n	Euler's Result
$f = x^2 + xy + y^2$	2	$x \cdot f_x + y \cdot f_y = 2f$
$f = x^3 + y^3$	3	$x \cdot f_x + y \cdot f_y = 3f$
$f = (x+y)/(x^2+y^2)$	-1	$x \cdot f_x + y \cdot f_y = -f$

7. DIRECTIONAL DERIVATIVE & GRADIENT

7.1 Gradient Vector

$$\nabla f = \left(\frac{\partial f}{\partial x}\right) \hat{i} + \left(\frac{\partial f}{\partial y}\right) \hat{j}$$

THREE VARIABLES

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

- ∇f points in direction of **maximum increase**
- $|\nabla f|$ = magnitude of maximum rate of change
- $\nabla f \perp$ level curves of f

7.2 Directional Derivative

$$D_{\hat{u}}f = \nabla f \cdot \hat{u}$$

IN DIRECTION OF UNIT VECTOR $\hat{u} = (A, B)$

$$D_{\hat{u}}f = f_x \cdot a + f_y \cdot b$$

where $a^2 + b^2 = 1$ (unit vector condition)

IF DIRECTION GIVEN AS ANGLE θ

$$\hat{u} = (\cos \theta, \sin \theta)$$

$$D_{\hat{u}}f = f_x \cos \theta + f_y \sin \theta$$

7.3 Key Results

Property	Formula / Result
Maximum directional derivative	$= \nabla f $ (in direction of ∇f)
Minimum directional derivative	$= - \nabla f $ (opposite to ∇f)
Zero directional derivative	When $\hat{u} \perp \nabla f$
Normal to surface $F(x,y,z) = c$	$\hat{n} = \nabla F / \nabla F $

8. TAYLOR & MACLAURIN SERIES (TWO VARIABLES)

8.1 Taylor Series of $f(x,y)$ about (a,b)

TAYLOR EXPANSION

$$\begin{aligned} f(x,y) = & f(a,b) + [(x-a)f_x + (y-b)f_y]_{(a,b)} \\ & + (1/2!) [(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}]_{(a,b)} \\ & + (1/3!) [\dots \text{third order terms} \dots] + \dots \end{aligned}$$

8.2 Maclaurin Series — Expansion about $(0,0)$

MACLAURIN SERIES ($A=0, B=0$)

$$\begin{aligned} f(x,y) = & f(0,0) + [x \cdot f_x(0,0) + y \cdot f_y(0,0)] \\ & + (1/2!)[x^2 f_{xx} + 2xy \cdot f_{xy} + y^2 f_{yy}]_{(0,0)} \\ & + (1/3!)[x^3 f_{xxx} + 3x^2 y \cdot f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy}]_{(0,0)} + \dots \end{aligned}$$

8.3 Compact Operator Form

$$f(a+h, b+k) = \sum (1/n!) (h \cdot \partial/\partial x + k \cdot \partial/\partial y)^n f(a,b)$$

📌 First-Order Terms

$$f(0,0) + x \cdot f_x + y \cdot f_y$$

Linear approximation

📌 Second-Order Terms

$$+\frac{1}{2}(x^2 f_{xx} + 2xy \cdot f_{xy} + y^2 f_{yy})$$

Quadratic approximation

9. JACOBIANS

9.1 Jacobian of Two Variables

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix}$$

EXPANDED FORM

$$J = (\partial u / \partial x)(\partial v / \partial y) - (\partial u / \partial y)(\partial v / \partial x)$$

9.2 Jacobian of Three Variables

$$J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{vmatrix}$$

9.3 Properties of Jacobians

Property	Formula	Remark
Chain Rule for Jacobians	$\frac{\partial(u,v)}{\partial(x,y)} = \left[\frac{\partial(u,v)}{\partial(r,s)} \right] \cdot \left[\frac{\partial(r,s)}{\partial(x,y)} \right]$	Jacobians multiply like derivatives
Inverse Jacobian	$J \cdot J' = 1 \quad \text{or} \quad \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$	$J' = 1/J$
Functional Independence	$J \neq 0$	Functions are independent
Functional Dependence	$J = 0$	Functions are dependent
Area Transformation	$dA_{uv} = J \cdot dA_{xy}$	Change of variables in integrals

9.4 Standard Jacobians

Transformation	Variables	Jacobian $ J $
Polar → Cartesian	$x = r \cos \theta, y = r \sin \theta$	r
Cylindrical	$x = r \cos \theta, y = r \sin \theta, z = z$	r
Spherical	$x = \rho \sin \varphi \cos \theta, y = \rho \sin \varphi \sin \theta, z = \rho \cos \varphi$	$\rho^2 \sin \varphi$

10. FUNCTIONAL DEPENDENCE

DEFINITION

CONDITION FOR INDEPENDENCE

$$J = \frac{\partial(u,v)}{\partial(x,y)} \neq 0$$

Functions $u = f(x, y)$ and $v = g(x, y)$ are **functionally dependent** if there exists a relation $\varphi(u, v) = 0$

CONDITION FOR DEPENDENCE

$$J = \partial(u, v) / \partial(x, y) = 0$$

Simple Meaning

- If $J = 0$: one function can be written in terms of others → **DEPENDENT**
- If $J \neq 0$: functions carry independent information → **INDEPENDENT**

EXAM APPROACH: TESTING FUNCTIONAL DEPENDENCE

- ① Compute u and v
- ② Find all 4 partial derivatives
- ③ Form the 2×2 Jacobian
- ④ Compute determinant
- ⑤ If $\det = 0 \rightarrow$ Dependent; find $\varphi(u, v) = 0$

11. MAXIMA & MINIMA (TWO VARIABLES)

11.1 Critical Points (First Order Condition)

$$f_x(x, y) = 0 \quad \text{AND} \quad f_y(x, y) = 0$$

★ Solve simultaneously to find critical point (a,b)

11.2 Second Order Test — Discriminant D

$$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$$

★ Evaluate D at the critical point (a, b)

$$D > 0 \ \& \ f_{xx} > 0$$

 **LOCAL MINIMUM**
f(a,b) is minimum value

$$D > 0 \ \& \ f_{xx} < 0$$

 **LOCAL MAXIMUM**
f(a,b) is maximum value

$$D < 0$$

 **SADDLE POINT**
Neither max nor min

$$D = 0$$

 **INCONCLUSIVE**
Higher order test needed

Alternative: Use f_{yy} instead of f_{xx}

$D > 0 \ \& \ f_{yy} > 0 \rightarrow \text{Min} \mid D > 0 \ \& \ f_{yy} < 0 \rightarrow \text{Max}$

11.3 Summary Table

Condition	Type of Point	Physical Meaning
$f_x=0, f_y=0, D>0, f_{xx}>0$	Local Minimum	Bowl shaped upward
$f_x=0, f_y=0, D>0, f_{xx}<0$	Local Maximum	Bowl shaped downward
$f_x=0, f_y=0, D<0$	Saddle Point	Max in one dir, min in another
$f_x=0, f_y=0, D=0$	Inconclusive	Need higher order analysis

12. LAGRANGE MULTIPLIERS

12.1 Problem Setup

OPTIMIZE

Maximize/Minimize: $f(x,y)$ – objective function

Subject to: $g(x,y) = 0$ – constraint equation

12.2 Core Condition

$$\nabla f = \lambda \cdot \nabla g$$

12.3 System of Equations to Solve

$$\partial f / \partial x = \lambda \cdot \partial g / \partial x$$

$$\partial f / \partial y = \lambda \cdot \partial g / \partial y$$

$$g(x, y) = 0 \quad \leftarrow \text{(Constraint)}$$

★ λ = Lagrange Multiplier (scalar) ★ Solve all 3 equations simultaneously for x, y, λ

12.4 Three Variables

OPTIMIZE $F(X, Y, Z)$ SUBJECT TO $G(X, Y, Z) = 0$

$$f_x = \lambda g_x \quad f_y = \lambda g_y \quad f_z = \lambda g_z \quad g(x, y, z) = 0 \rightarrow \text{solve for } x, y, z, \lambda$$

LAGRANGE METHOD — STEP-BY-STEP

① Write $F(x, y, \lambda) = f(x, y) - \lambda \cdot g(x, y)$ ② Set $\partial F / \partial x = 0, \partial F / \partial y = 0, \partial F / \partial \lambda = 0$ ③ Solve system $\rightarrow$ critical points ④ Evaluate f at each point to find max/min

Q1: What is a partial derivative $\partial f/\partial x$?

A: Rate of change of f w.r.t. x keeping all other variables **constant**

Q2: State the total differential of $z = f(x,y)$

A: $dz = (\partial z/\partial x)dx + (\partial z/\partial y)dy$

Q3: Condition for functional dependence of u,v ?

A: $J = \partial(u,v)/\partial(x,y) = 0$

Q4: What is the gradient vector ∇f ?

A: $\nabla f = (\partial f/\partial x)\hat{i} + (\partial f/\partial y)\hat{j}$ – vector of all partial derivatives

Q5: Directional derivative formula?

A: $D_{\hat{u}}f = \nabla f \cdot \hat{u}$ where $\hat{u}$ is a **unit vector**

Q6: What does $D < 0$ mean in max/min test?

A: $D < 0 \rightarrow$ **Saddle Point** (neither max nor min)

Q7: Euler's theorem for homogeneous function?

A: $x \cdot f_x + y \cdot f_y = n \cdot f$ where n = degree of homogeneity

Q8: What is the discriminant D in max/min?

A: $D = f_{xx} \cdot f_{yy} - (f_{xy})^2$

Q9: Clairaut's theorem states?

A: If f_{xy} and f_{yx} are continuous $\rightarrow f_{xy} = f_{yx}$ (order of differentiation doesn't matter)

Q10: Lagrange multiplier condition?

A: $\nabla f = \lambda \nabla g$ (gradients are parallel at optimal point)

Q11: Jacobian for polar transformation?

A: $x = r \cos\theta, y = r \sin\theta \rightarrow J = r$

Q12: Condition for local minimum?

A: $f_x = 0, f_y = 0, D > 0$, and $f_{xx} > 0$

Q13: Chain rule for $z=f(x,y), x=x(t), y=y(t)$?

A: $dz/dt = (\partial z/\partial x)(dx/dt) + (\partial z/\partial y)(dy/dt)$

Q14: Maclaurin series expands about which point?

A: About **(0, 0)** – special case of Taylor series at origin

Q15: Direction of maximum rate of increase?

A: Along ∇f ; maximum value = $|\nabla f|$

Q16: Implicit differentiation for $F(x,y)=0$?

A: $dy/dx = -F_x/F_y$ (provided $F_y \neq 0$)

Q17: What is the inverse Jacobian property?

A: $\partial(u,v)/\partial(x,y) \cdot \partial(x,y)/\partial(u,v) = 1$

Q18: Continuity conditions in two variables?

A: ① $f(a,b)$ defined ② limit exists ③ limit = $f(a,b)$

Q19: What does $D = 0$ mean in the second derivative test?

A: Test is **inconclusive** – use other methods (higher order analysis)

Q20: Lagrange: how many equations to solve?

A: $n+1$ equations for n variables + 1 constraint (includes the constraint itself)

⚡ EXAM TIPS & COMMON MISTAKES

✗ **MISTAKE 1: Partial Derivative**
Wrong: Differentiating y-terms when finding $\partial/\partial x$
Right: Treat ALL other variables as **constants** when differentiating w.r.t. one variable

✗ **MISTAKE 2: Chain Rule**
Wrong: Forgetting intermediate variable terms
Right: Include ALL paths. For $z = f(x,y)$, $x=x(t)$, $y=y(t)$:
 $dz/dt = (\partial z/\partial x)(dx/dt) + (\partial z/\partial y)(dy/dt)$ — **both** terms needed

✗ **MISTAKE 3: Jacobian Sign**
Wrong: Getting signs wrong in 2×2 determinant
Right: $J = ad - bc$ for matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
Always double-check: $(\partial u/\partial x)(\partial v/\partial y) - (\partial u/\partial y)(\partial v/\partial x)$

✗ **MISTAKE 4: Saddle vs Max/Min**
Wrong: Declaring max/min without checking D
Right: Always compute $D = f_{xx}f_{yy} - (f_{xy})^2$ first.
 $D < 0$ is **always** a saddle point regardless of f_{xx}

✗ **MISTAKE 5: Non-Unit Vector**
Wrong: Using any vector in directional derivative
Right: ALWAYS convert to unit vector first:
 $\hat{u} = v/|v|$ before computing $D_{\hat{u}}f = \nabla f \cdot \hat{u}$

✗ **MISTAKE 6: Euler's Theorem Degree**
Wrong: Wrong degree for non-polynomial functions
Right: Verify $f(tx,ty) = t^n \cdot f(x,y)$ to confirm degree.
Rational functions can have negative degrees!

✗ **MISTAKE 7: Lagrange Setup**
Wrong: Not including ALL equations from $\nabla f = \lambda \nabla g$
Right: You need $f_x = \lambda g_x$, $f_y = \lambda g_y$, AND $g = 0$
Three equations for three unknowns (x, y, λ)

✗ **MISTAKE 8: Limit Along Different Paths**
Wrong: Checking limit along only one path
Right: To show limit DNE: find 2 paths giving different values.
Common paths: $y=0$, $x=0$, $y=x$, $y=mx$, $y=x^2$

✓ SCORING STRATEGIES

Topic	High-Yield Exam Tips	Marks Trap
Partial Derivatives	Show step clearly; treat other vars as constants	Forgetting product/chain rule within
Maxima & Minima	Always compute D; classify clearly	Not checking second derivative test
Lagrange	Set up all 3 equations; eliminate λ first	Skipping constraint equation
Jacobian	Write 2×2 matrix clearly; expand determinant	Sign errors in determinant expansion
Chain Rule	Draw dependency diagram before computing	Missing intermediate variable paths
Euler's Theorem	Identify degree first; verify with substitution	Wrong degree for composite functions

🏆 **TOP 5 GUARANTEED MARKS TIPS**

- **Show critical point computation** — partial marks given
- **State theorems before applying** — earns method marks

- **Verify unit vector** — confirm $a^2+b^2=1$ in directional derivative
- **Write $J = 0$ / $J \neq 0$ conclusion** explicitly in Jacobian problems

- **Box your final answer** — examiner sees it instantly

⚡ NO EXPLANATIONS — PURE FORMULAS — LAST MINUTE REVISION ⚡

 PARTIAL DERIVATIVES

$\partial f / \partial x$: treat $y = \text{const}$
 $\partial f / \partial y$: treat $x = \text{const}$
 $f_{x_y} = f_{y_x}$ (Clairaut)

 TOTAL DIFFERENTIAL

$dz = f_x dx + f_y dy$
 $dz/dt = f_x \dot{x} + f_y \dot{y}$

 CHAIN RULE

$\partial z / \partial s = z_x \cdot x_s + z_y \cdot y_s$
 $\partial z / \partial t = z_x \cdot x_t + z_y \cdot y_t$
 $dy/dx = -F_x / F_y$

 GRADIENT & DIRECTIONAL

$\nabla f = f_x \hat{i} + f_y \hat{j}$
 $D_{\hat{u}} f = \nabla f \cdot \hat{u}$
 Max $D = |\nabla f|$
 Min $D = -|\nabla f|$

 EULER'S THEOREM

$x \cdot f_x + y \cdot f_y = n \cdot f$
 $x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1)f$

 TAYLOR (ABOUT (A,B))

$f(a,b) + (h \cdot f_x + k \cdot f_y) + (1/2!)(h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \dots$ [$h=x-a, k=y-b$]

 JACOBIAN

$J = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix}$
 $J \cdot J' = 1$
 Polar: $J = r$
 Spherical: $J = \rho^2 \sin \varphi$

 MAXIMA & MINIMA

$f_x = 0, f_y = 0$
 $D = f_{xx} f_{yy} - (f_{xy})^2$
 $D > 0, f_{xx} > 0 \rightarrow \text{Min}$
 $D > 0, f_{xx} < 0 \rightarrow \text{Max}$
 $D < 0 \rightarrow \text{Saddle}$

 LAGRANGE

$\nabla f = \lambda \cdot \nabla g$
 $f_x = \lambda g_x, f_y = \lambda g_y$
 $g(x,y) = 0$

 FUNCTIONAL DEPENDENCE QUICK TEST

$J = 0 \rightarrow \text{DEPENDENT} \checkmark$

$J \neq 0 \rightarrow \text{INDEPENDENT} \checkmark$

 CONTINUITY QUICK CHECK

① $f(a,b)$ exists

② $\lim f(x,y)$ exists

③ $\lim = f(a,b)$

 SYMBOL REFERENCE

Symbol	Meaning	Symbol	Meaning	Symbol	Meaning
∂	Partial derivative	∇	Gradient (nabla)	λ	Lagrange multiplier
J	Jacobian	D	Discriminant	$\hat{u}$	Unit vector
f_x	$\partial f / \partial x$	f_{xx}	$\partial^2 f / \partial x^2$	f_{xy}	$\partial^2 f / \partial y \partial x$

